



DBMS 01 | Introduction Of RDBMS | CS & IT | GATE 2024 FastTrack Batch

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Summary

AI Notes

AI Slides

Text Notes

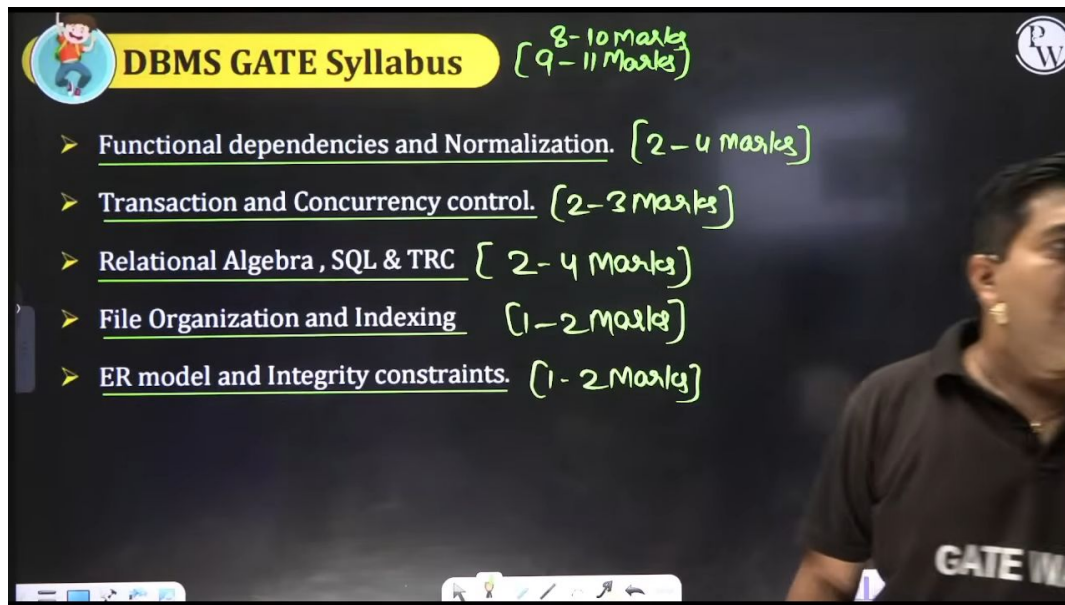
Screenshots

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▶ 4:13



Introduction of DBMS

Data : Facts (Raw Material)

↓

Information (Meaningful Data)

↓

Database { Collection of Logically Related / Inter-Related Data
Collection of similar Records }

↓

DBMS { Set of Program (S/W) Used to Access & Update the DB in efficient & fast manner }

▶ 17:39



Topics : Introduction of RDBMS

RDBMS → Relation (Table)

Relation (Table) → Row

Relation (Table) → Column

Row : Tuple / Record

Column : field / Attribute

(Degree) **Arity** : # of Attributes / fields

Cardinality : # of Tuples / Records

STUDENT

Roll No	Name	Branch	CGPA
1	A	CS	9
2	B	IT	10
3	C	CS	9
4	D	IT	10
5	E	EC	9
6	F	CS	10
7	G	IT	9

▶ 23:23

Relation Schema : Table Abstraction (Table Heading)

- Table (Relation) Name
- Attribute Name
- Attribute Domain (Data type)

STUDENT (^{Integer}RollNo, ^{String}Name, ^{String}Branch, ^{Real}CGPA)

Relation Instance : Set of Records

▶ 31:40

Domain

- The set of allowed values for each attributes is called the domain in the attribute.

- Roll # : alphanumeric string
- First Name, Last Name : Alpha String
- DoB : Date
- Passport # : String (Letter followed by 7 digits)-nullable
- Aadhar # : 12-digit number
- Department : Alpha String

Attribute values are (normally) required to be atomic; that is, indivisible.

The special value null is a member of every domain. Indicated that the value is

▶ 35:04



Relation Schema and Instance



- $A_1, A_2, \dots, A_n$ are attributes
- $R = (A_1, A_2, \dots, A_n)$ is a relation schema

Example:

instructor = (ID, name, dept_name, salary)

- Formally, given sets $D_1, D_2, \dots, D_n$ a **relation r is a subset of**
 $D_1 \times D_2 \times \dots \times D_n$

Thus, a relation is a set of n -tuples $(a_1, a_2, \dots, a_n)$ where each $a_i \in D_i$

- The current values (**relation instance**) of a relation are specified by a table.
- An element t of r is a tuple, represented by a row in a table.

▶ 35:37



Relation Schema and Instance



- A relation schema R , denoted by $R(A_1, A_2, \dots, A_n)$, is made up of a relation name R , a list of attributes, $A_1, A_2, \dots, A_n$ & Each attribute A_i domain D_i . D_i is called the domain of A_i and is denoted by $\text{dom}(A_i)$.
- The **degree (or arity)** of a relation is the number of attributes n of its relation schema.
- STUDENT(Name, Ssn, Home_phone, Address, Office_phone, Age, Gpa)
- A relation of degree seven, which stores information about university students, would contain seven attributes describing each student as follows:

Using the data type of each attribute, the definition is sometimes written as:

STUDENT(Name: string, Ssn: string, Home_phone: string, Address: string, Office_phone: string, Age: integer, Gpa: real)

▶ 43:29

Relation intension and Relation extension

- A relation (or **relation state**) r of the relation schema $R(A_1, A_2, \dots, A_n)$, also denoted by $r(R)$, is a set of n -tuples $r = \{t_1, t_2, \dots, t_m\}$.

The terms **relation intension** for the **schema R** and **relation extension** for a relation state $r(R)$ are also commonly used.

Relation State (relation extension)
Relation intension : Schema
Relational Instance

▶ 54:25

Why this model is called Relational Model..

- The definition of a relation can be restated more formally using **set theory concepts** as follows.
- A relation (or relation state) $r(R)$ is a **mathematical relation** of degree n on the domains $\text{dom}(A_1), \text{dom}(A_2), \dots, \text{dom}(A_n)$, which is a subset of the Cartesian product (denoted by $\times$) of the domains that define R :

$$r(R) \subseteq (\text{dom}(A_1) \times \text{dom}(A_2) \times \dots \times \text{dom}(A_n))$$

- The Cartesian product specifies all possible combinations of values from the underlying domains. Hence, if we denote the total number of values, or **cardinality**, in a domain D by $|D|$ (assuming that all domains are finite), **the total number of tuples in the Cartesian product is**

$$|\text{dom}(A_1)| \times |\text{dom}(A_2)| \times \dots \times |\text{dom}(A_n)|$$

▶ 1:01:31



Why this model is called Relational Model..

- This product of cardinalities of all domains represents the total number of possible instances or tuples that can ever exist in any relation state $r(R)$.
- Out Of all these possible combinations, a relation state at a given **time-the current relation state**-reflects only the valid tuples that represent a particular state of the real world.
- In general, as the state of the real world changes, so does the relation state, by being transformed into another relation state.

▶ 1:04:11



Topics: Functional Dependency $[x \rightarrow y]$

Functional Dependency $[x \rightarrow y]$

Let R be the Relational Schema, x & y be the attribute Set of Relation R . & t_1 & t_2 Any Two Tuples such that

$x \rightarrow y$ iff:

If $t_1.x = t_2.x$ then $t_1.y = t_2.y$ Must be Same.

▶ 1:13:28

 **Topics: Functional Dependency**

determinant
 $X \rightarrow Y$
determines / dependent

X functionally determines Y
or
 Y is functionally dependent on X .

▶ 1:21:58

 **Topics: Functional Dependency**

Type of FD :

- ① Trivial FD
- ② Non Trivial FD
- X ③ Semi Non Trivial FD.

▶ 1:23:55

Topics: Functional Dependency

① Trivial FD :

$X \rightarrow Y$ is Trivial FD

iff $X \supseteq Y$

R.H.S Attributes Must be Part or equals to L.H.S Attributes.

eg

- $AB \rightarrow A$
- $AB \rightarrow B$
- $AB \rightarrow AB$
- Roll No Name $\rightarrow$ Roll No
- Roll No Name $\rightarrow$ Name
- Roll No Name $\rightarrow$ Roll No Name

▶ 1:26:35

Topics: Functional Dependency

② Non Trivial FD :

$X \rightarrow Y$ is Non Trivial FD

iff $X \cap Y = \phi$ & $X \rightarrow Y$ must satisfy FD Definition.

eg

- $A \rightarrow B$
- $A \rightarrow C$
- Sid $\rightarrow$ Sname

▶ 1:30:07



③ semi Non Trivial FD

$X \rightarrow Y$ is Semi Non Trivial FD.

iff $X \supsetneq Y$ & $X \cap Y \neq \emptyset$

eg $AB \rightarrow BC$

▶ 1:33:33

FD. $X \rightarrow Y$

If $t_1.X = t_2.X$ then $t_1.Y = t_2.Y$ Must be Same.

Whenever in $X \rightarrow Y$, X value Repeat then Corresponding Y value Must be Same.

(Note)

Trivial FDs are always Valid.

▶ 1:36:04

Important Point about FD's

- With the Using of Relation Instance (Only from Particular Instance (Relation State (extension)) we can Conclude Only that FD which is Not Valid (does not hold)

By Particular Instance (State (extension)) we can not conclude that FD (Any FD) ^{always} Hold Bcz State (extension) can be change Bcz FD is Property of Relational Schema.

▶ 2:08:25

[MCQ]

[GATE:2 Marks]



#Q. From the following instance of a relation scheme R (A, B, C), we can conclude that:

A	B	C
1	1	1
1	1	0
2	3	2
2	3	2

- A** A functionally determines B and B functionally determines C
- B** A functionally determines B and B does not functionally
- C** B does not functionally determines C
- D** A does not functionally determines B and B does not fu

▶ 2:12:05

[MCQ]

[GATE:2 Marks]



#Q. Given the following relation instance.

X	Y	Z
1	4	2
1	5	3
1	6	3
3	2	2

Which of the following functional dependencies are satisfied by the instance?

A $XY \rightarrow Z$ and $Z \rightarrow Y$

B $YZ \rightarrow X$ and $Y \rightarrow Z$

C $YZ \rightarrow X$ and $X \rightarrow Z$

D $XZ \rightarrow Y$ and $Y \rightarrow X$

▶ 2:16:12

Armstrong's Axioms/Inference Rules



- ❑ Axioms, or rules of inference, provide a simpler technique for reasoning about functional dependencies
- ❑ In the rules that follow, we use Greek letters ($\alpha, \beta, \gamma, \dots$) for sets of attributes.
- ❖ We can use the following three rules to find logically implied functional dependencies.
- ❖ By applying these rules repeatedly, we can find all of F^+ , given F . This collection of rules called Armstrong's Axioms in honor of the person who first proposed it.
 - **Reflexivity Rule:** If α is a set of attributes and $\beta \subseteq \alpha$, then $\alpha \rightarrow \beta$ holds.
 - **Augmentation rule:** If $\alpha \rightarrow \beta$ holds and γ is a set of attributes, then $\gamma\alpha \rightarrow \gamma\beta$ holds.
 - **Transitivity Rule:** If $\alpha \rightarrow \beta$ holds and $\beta \rightarrow \gamma$, then $\alpha \rightarrow \gamma$ holds.

▶ 2:20:37



Topics: Attribute closure $[X]^+$



Attribute closure $[X]^+$: Let R be the Relational Schema with attribute set X .

Set of All possible Attributes which is logically/ functionally determined by attribute X **Called**

▶ 2:28:21